Ride-Hailing Demand & Smart Charging for an Electric Fleet

Advanced Analytics and Applications

Team 04

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1 Problem description

Urban mobility is in the middle of a structural shift. Over the past decade, ride-hailing has grown from a niche offering into a force that is disruptively reshaping urban transport (Wang & Yang, 2019), changing travel behavior and, in a number of cities, drawing riders away from public transit (Tirachini, 2020). At the same time, road transport is electrifying rapidly: extrapolating historical sales with a logistic growth model, electric vehicles are projected to reach roughly a third of the global passenger-car stock by the early 2030s (Rietmann et al., 2020). These two trends are converging, and established car manufacturers increasingly seek to move beyond selling vehicles towards operating mobility services themselves. This is where our **business goal** begins: a German car company is entering the space with a ride-hailing platform run on a fully electrified fleet, targeting the more receptive US market. Building vehicles and profitably operating a fleet of them are distinct competencies, and running an on-demand service requires a quantitative understanding of where and when demand arises, which the manufacturer does not yet possess.

That understanding matters because demand for on-demand mobility has pronounced spatial and temporal structure (Ke et al., 2017) and is further shaped by exogenous conditions such as weather. An operator must therefore decide on three levels. Strategically, which districts of a city to enter first; tactically, how many vehicles to deploy and at which times and places; and operationally, how to position and, since the fleet is electric, charge those vehicles efficiently. Answering these questions requires demand data at fine spatial and temporal resolution, which does not yet exist for a platform that has not launched. As a proxy, we use the public taxi trip records of the City of Chicago from 2024 onwards: taxis are a reasonable stand-in for on-demand ride-hailing, and Chicago serves as a representative large US city. Nevertheless, the proxy is imperfect, as taxis differ from app-based hailing in booking behavior and market coverage, a limitation we keep in view when drawing conclusions.

From this business goal we derive our **data-mining goal**, a chain of five analytics tasks that build on one another. First, *data preparation* turns the large, privacy-restricted raw trip records into demand panels: we validate and clean the records, discarding implausible or inconsistent trips, discretize the city with a hexagonal H3 grid, and enrich each cell and time window with external weather data. Second, *descriptive spatial analytics* characterizes how demand varies across space and time, selects an appropriate spatial and temporal resolution, and identifies persistent demand hotspots using Gaussian Mixture Models and kernel density estimation. Third, *predictive analytics* estimates the expected demand per spatial unit and time window as a function of calendar, weather and location, using Support Vector Machines and feed-forward neural networks. The aim is conditional estimation, quantifying how demand depends on observable conditions, rather than autoregressive forecasting of the next interval from the current one. Fourth, *prescriptive analytics* addresses the electric dimension by designing a home-charging agent through reinforcement learning that minimizes recharging cost while ensuring enough energy for the working day. Finally, *synthesis* consolidates these results into recommendations for the operator, including whether it should rely on private or public charging infrastructure. Together, the five tasks move from describing demand, to explaining it, to acting on it.

2 Data description

We draw on three datasets. The primary source is the *Taxi Trips (2024-)* record published by the City of Chicago on its open data portal, holding 14,802,849 trips across 23 fields. It spans 1 January 2024 to 31 March 2026 and covers 3,735 vehicles, 46 licensed taxi companies and all 77 of Chicago's community areas. To this we add historical weather for Chicago from the Open-Meteo API, at hourly resolution (19,728 observations of temperature, precipitation, rain, snowfall, wind speed and a categorical WMO weather code) alongside 822 daily aggregates that add sunshine duration and the number of precipitation hours. A GeoJSON of the 77 community-area boundaries supplies the geometry used to map and aggregate trips spatially. We decided against point-of-interest extensions, as the records already at hand cover the questions set out in Section 1, so the expected analytical gain did not justify the additional integration effort.

Each row represents one completed taxi trip. Location is reported at three nested levels, census tract within community area, plus a centroid coordinate pair. The 23 fields group as follows, with the full field-by-field dictionary given in the Appendix. For demand modeling, only the pickup-side time and location fields are required.

Table 1: Structure of the raw Chicago taxi trip records, with the 23 fields grouped by type.

Group	**Fields**	**Notes**
Identifiers	Trip ID, Taxi ID, Company	3,735 vehicles, 46 companies
Time	Trip Start / Trip End Timestamp	rounded to 15-minute bins
Trip metrics	Trip Seconds, Trip Miles	duration and distance
Space	Pickup / Dropoff Census Tract, Community Area, Centroid Latitude, Longitude and Location	coordinates are area centroids, not exact points
Money	Fare, Tips, Tolls, Extras, Trip Total	0.21% missing
Payment	Payment Type	

Two structural properties shape everything that follows. First, all timestamps are rounded to quarter-hour bins, taking only the minute values 0, 15, 30 and 45, which sets a hard floor on the temporal resolution any analysis can achieve. Second, location is restricted for **privacy** rather than by measurement: no exact coordinates are published, and the coordinate fields give the centroid of the enclosing area. Tract identifiers are further suppressed for the majority of trips, missing in 55.65% of pickups and 56.98% of dropoffs, whereas the community area is far more complete, missing in 2.79% and 8.92% of cases. Because suppression applies wherever a combination of tract and time holds too few trips to guarantee anonymity, it removes precisely the sparse observations and is therefore correlated with demand density. Analysis at tract level would be biased towards dense areas, which is why the choice of spatial unit in the next section cannot simply default to the finest available geography.

A final limitation is conceptual rather than technical. The dataset records completed trips and therefore captures **served demand** only: requests that went unmet leave no trace, so the data understate true demand in exactly those places and periods where supply was scarce.

3 Data preparation

Turning the raw records into an analyzable demand panel proceeds in three stages: cleaning, discretization and aggregation. Syntactic cleaning harmonizes data types and parses timestamps, and a duplicate check confirms that no fully duplicated records exist. Semantic validation then distinguishes two kinds of implausibility, and that distinction carries the whole preparation. Records that cannot describe a real journey are excluded outright: 98 trips end before they start, 12,104 last longer than three hours, and 1,202 exceed 100 miles. Records implausible in only one attribute are retained as demand events but withheld from the statistics they would distort, namely 1,294,820 trips of zero distance, 201,927 of zero duration, and fares above 200 dollars. A zero-distance record still marks a real pickup: deleting it would understate demand, while averaging it would corrupt the distance statistics, so it counts towards demand and is suppressed only where it would mislead. The complete set of validation rules and thresholds is documented in the Appendix.

Missing values are likewise not treated globally. Since the demand model describes where trips begin, missing dropoff information never removes an observation. Each derived dataset carries its own requirement, so every analysis runs on the largest subset valid for it: the hexagon panel needs pickup coordinates, the community-area panel needs the community area. Requiring pickup coordinates removes 405,819 trips, which together with the hard exclusions leaves 14,382,862 of 14,802,849 records, or 97.2%. We checked whether this distorts the temporal picture. Across months the missing rate is stable between 2.4% and 3.4%, but across the day it is not: it falls to 1.9% in the late afternoon and rises to 6.8% around 4 a.m. This is the density-driven mechanism from Section 2, as fewer night-time trips push more area and time combinations below the anonymity threshold. The effect is small, but night-time demand is marginally understated.

For the spatial **resolution** we compared Chicago's community areas with a hexagonal grid. Community areas are human-readable and therefore valuable for interpretation, but they vary considerably in size and shape, so demand densities are not comparable across them. We therefore discretize the city with the H3 indexing system, developed by Uber for this purpose, which provides cells of uniform size and shape at a tunable resolution r4 to r8, where a higher index means smaller cells and each step divides the area by roughly seven (Table 7 gives an intuitive scale for each). Two independent criteria bound the sensible range from both sides, as Table 2 shows. Downwards, r4 and r5 leave only 4 and 8 cells for the entire city, three of which already carry 80% of all demand, far too coarse to guide where vehicles should be deployed. Upwards, two things break at once. Sparsity rises steeply, with the share of empty cell and window combinations climbing from 12.6% at r6 to 49.2% at r7 and 82.0% at r8, so the panel increasingly records absence rather than demand. More fundamentally, measurement validity fails: because coordinates are area centroids of uneven precision, with 607 tract centroids for trips of known tract and only 77 community-area centroids for suppressed ones, a hexagon smaller than its reporting area cannot observe real structure. At r8, 67% of occupied hexagons are fed by a single centroid, so their apparent detail only reproduces where centroids happen to lie. Resolution r6 is the finest level at which both criteria remain intact, at more than three times the detail of r5, and we adopt it as canonical.

Table 2: Spatial resolution sweep at fixed four-hour windows.

Resolution	Cell area	Cells	Empty share	Single-centroid cells
r4	1,770.3 km^2	4	1.8%	0%
r5	252.9 km^2	8	1.0%	0%
r6	36.1 km^2	27	12.6%	0%
r7	5.2 km^2	128	49.2%	20%
r8	0.7 km^2	475	82.0%	67%

Quarter-hour rounding of the source timestamps bounds the temporal grid from below. We build one-hour, four-hour, eight-hour and daily windows and adopt four hours as canonical, since hourly bins reproduce the same sparsity problem while daily bins average away the within-day pattern on which any deployment decision depends. The weather records are aggregated to these windows and joined on the time key, contributing temperature, rain, snowfall, total precipitation, wind speed and the categorical weather code, so that demand variation driven by conditions is not misattributed to the calendar.

The **aggregation** returns one row per hexagon and time window, carrying the number of pickups as the target variable alongside mean trip distance, duration and total fare amount, and the number of distinct taxis active in the cell. The window timestamp is retained as the basis for the cyclical calendar features, namely time of day, day of week and month, that enter the models in Section 4. Separately, we derive idle time per vehicle as the gap between one trip ending and the same taxi starting its next, which supports the utilization view in Section 4. One step deserves emphasis. A grouped aggregation produces rows only where at least one trip occurred, so a cell without pickups is absent rather than zero, and a model trained on that panel would never observe the quiet half of the distribution and would systematically overestimate demand. We therefore complete the panel across the full Cartesian product of cells and windows and set the missing combinations to zero, leaving trip characteristics undefined because no trips exist to average. The canonical dataset comprises 27 hexagons across 4,927 four-hour windows, that is 133,029 observations, of which 12.6% are genuine zeros introduced by this step alone. Figure 1 makes the trade-off visible.

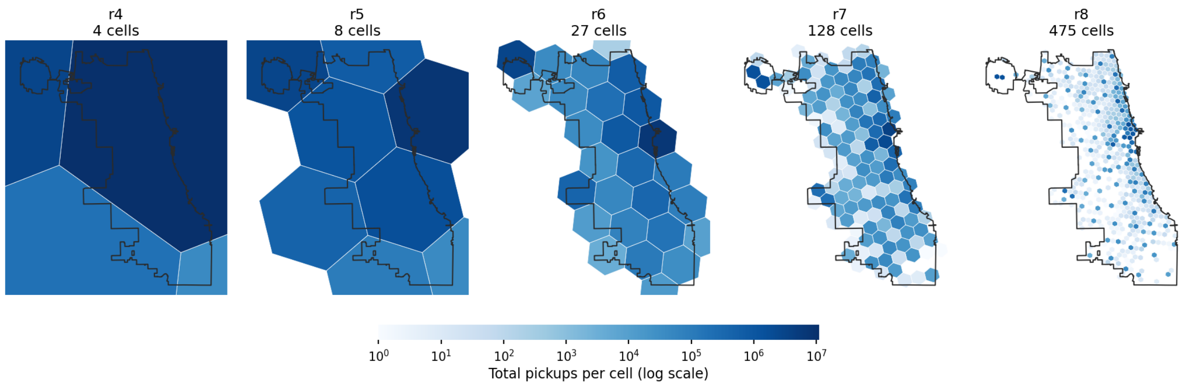

Figure 1: Total pickups per cell across H3 resolutions r4 to r8, on a shared logarithmic color scale, with the city boundary for orientation. Coarse grids collapse the city into a handful of cells, while at r7 and r8 the surface fragments into largely empty ones. Resolution r6 is the finest level that still resolves spatial variation without dissolving into noise.

4 Data analytics

4.1 Descriptive analytics

City-wide demand follows a stable and highly predictable rhythm. Weekdays are markedly busier than weekends, averaging 18,922 against 13,919 trips per day, and both peak in the late afternoon at 5 p.m. Their shapes differ in a way characteristic of commuting, as Figure 2 shows: on weekdays the hours from 6 to 9 a.m. carry 18.5% of the day's trips, against only 10.5% at weekends, while the weekend redistributes demand into the small hours, where midnight to 4 a.m. accounts for 6.8% of trips against 3.1% on weekdays (see also Figure 4). The pattern repeats across the year with a pronounced seasonal amplitude, rising from about 13,700 trips per day in January to 20,400 in June (see Figure 5). Weather adds a genuine secondary effect on top of this calendar structure. Demand is generally higher in warmer conditions, rising from roughly 2,400 trips per four-hour window in the coldest decile of observations to 4,142 in the warmest, and rain lifts demand by a consistent 5 to 7% (see Figure 6 and Figure 7). Neither association is an artifact of mild weather coinciding with busy hours, as both persist when windows are compared within the same time of day. Snowfall, by contrast, shows no reliable effect in our sample, since windows with heavy snow are too rare to support a conclusion (see Figure 8). We read these patterns as commuting producing the weekday morning shoulder, business and nightlife density sustaining the long afternoon plateau, leisure travel shifting the weekend into the night, and the weather response reflecting substitution away from walking and public transit in unpleasant conditions.

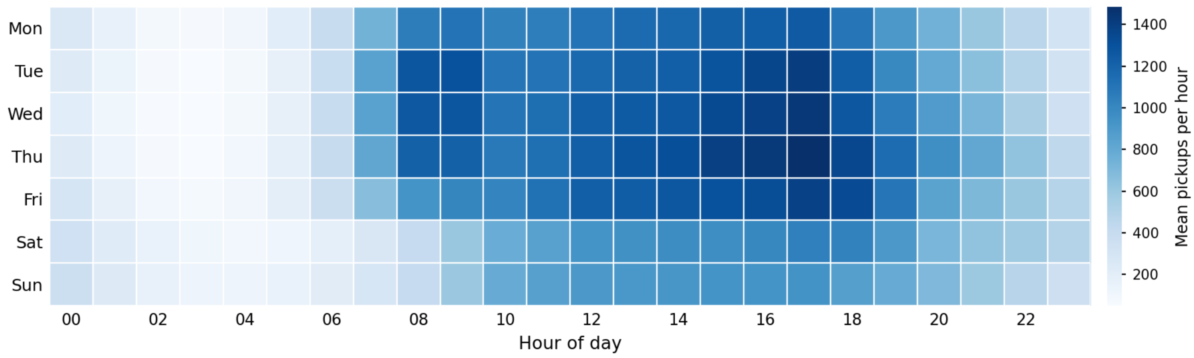

Figure 2: Mean pickups by weekday and hour of day. The weekday morning shoulder from 6 to 9 a.m. is absent at weekends, while weekend demand extends further into the night. Both patterns peak in the late afternoon.

Spatially, demand is extremely concentrated: at the canonical resolution a single downtown cell accounts for 46.8% of all pickups, and together with O'Hare for 67.5%. The more consequential finding is qualitative, as the two cores operate in different ways. The downtown core generates a high throughput of short trips, whereas the airports generate fewer but considerably longer and more valuable ones, with O'Hare averaging 14.2 miles per trip and the highest tip rate in the city at 19%. Origin-destination flows are asymmetric, so vehicles accumulate at destinations and must eventually be repositioned, and idle time is highest in peripheral areas and low-demand hours, suggesting local over-supply.

To test whether these cores are genuine rather than an artifact of our own grid, we identified hotspots without reference to any grid, fitting a Gaussian mixture model to the raw pickup coordinates and adding a kernel density estimate for the continuous density gradient (see Figure 9 and Figure 10). The mixture uses eight components, chosen for interpretability after the Bayesian information criterion continued to decline across the tested range. The component centers fall on the same cores, the Loop, Near North, O'Hare and Midway, confirming them as real features of the demand surface rather than products of our discretization. Comparing these grid-free centers with the leading cells of each hexagon resolution also shows how the grid distorts them: r4 merges the Loop and Near North into one cell, while r8 splits a single airport across several. Resolution r6 aligns most closely with the mixture centers, which corroborates the choice made in Section 3 on entirely independent grounds.

4.2 Predictive analytics

We now estimate demand for each spatial unit and time window. The models describe demand as a function of observable conditions rather than extrapolating it from recent demand, because the operational question is what demand to expect under given circumstances, not what follows mechanically from the preceding interval. The target is the pickup count per hexagon and four-hour window. It is heavily right-skewed and contains 12.6% zeros, so we model it on a log scale and invert the transformation before scoring. The eighteen predictors comprise the cyclical encodings of time of day, day of week and month, a binary weekend indicator, the cell centroid coordinates, the great-circle distances to the three demand centers identified in Section

4.1, and six four-hourly weather variables. Representing location as continuous coordinates and distances rather than one indicator per cell allows both models to share information between neighboring units and to generalize across space, which is what a location-conditioned question requires. Full feature definitions are given in the Appendix.

The **validation strategy** is built around one specific leakage risk. The four-hour windows of a single day share weather, calendar values and unobserved local shocks, so a random split would place some windows of a day in training and the rest in test, letting a model exploit day-specific signal that its features do not contain. We therefore split by whole calendar days, so that a day appears entirely in training or entirely in test. A grouped random sample of 15% of days forms an untouched holdout, tuning uses grouped cross-validation on the remaining development set, and standardization, the target transform and the hotspot anchors are fitted on training data only and refitted within each fold. We report three measures. MAE gives the average error in trips and is the most interpretable, RMSE penalizes the large misses that occur at demand peaks, and R^2 is scale-free. Only R^2 remains comparable when the resolution changes, for reasons set out below, so we lead the comparison with it.

We begin with a linear support vector regressor as a kernel-free baseline, add an RBF kernel selected by grid search, and compare both against a feedforward neural network in a baseline and a tuned configuration. Table 3 reports holdout performance. The linear model explains 63% of the variance in demand, which is inadequate and expected, since it cannot represent the interaction between location and time of day that drives the pattern. The RBF kernel raises this to 88%, and the neural network to 93%, with tuning reaching 96% and roughly halving the average error in trips relative to the kernel machine. The network architecture, the tuning grid and the training diagnostics are documented in the Appendix (see Figure 11, Table 11, Figure 12 and Figure 13).

Table 3: Holdout performance at H3 r6 and four-hour windows.

Model	**R^2**	**MAE**	**RMSE**	**Inference**
Linear SVM	0.656	64.10	196.92	0.03 s
RBF SVM	0.875	30.83	118.91	29.18 s
Neural network, baseline	0.926	26.75	91.36	0.01 s
Neural network, tuned	0.957	19.35	69.93	0.01 s
Naive benchmark (historical mean)	0.898	29.14	107.13	

Whether this complexity is justified is decided by **benchmarking** against a trivial alternative. A lookup table that simply returns the historical mean demand for each cell and time of day, with no learning at all, reaches R^2 0.898. The RBF support vector machine, after a full grid search, reaches 0.875 and therefore fails to beat this lookup on either measure. Only the neural network adds genuine signal, reaching 0.957 and reducing the average error by a third relative to the benchmark. The computational comparison points the same way: the kernel machine needs 29 seconds to score the holdout, because kernel methods scale roughly quadratically in the number of training samples, whereas the network needs 0.01 seconds, a factor of about two thousand, and it trains on the full development set rather than a subsample. Interpretability is not the trade-off it is commonly assumed to be, since a SHAP analysis (see Figure 14) shows the distances to the demand centers and the time-of-day terms dominating the prediction, which is a defensible explanation for a non-technical audience. The deep-learning approach is therefore warranted here, not because it is more elaborate, but because it is the only model that improves on a lookup table while also being cheaper to serve.

Performance across resolutions must be read with care, because raw MAE falls mechanically as cells become finer and hold fewer trips. Census tract achieves an MAE of 1.5 against 62.5 at r5, yet is by far the weaker model, which is why the comparison rests on R^2. Refitting both models at each resolution exposes the granularity and sparsity trade-off directly. These refits reuse the tuned configuration rather than repeating the search at every resolution, so their values are slightly below those in Table 3 and should be read as a comparison across resolutions rather than against the holdout result. Spatially, performance degrades as the grid tightens: the network reaches R^2 0.945 at r5 and 0.942 at r6, falls to 0.775 at r7, where 49% of observations are empty, and collapses to 0.375 at census tract level. Community areas sit between r6 and r7 on every dimension, with 77 units against 27 and 128, a 17.3% zero rate against 12.6% and 49.2%, and R^2 0.922 against 0.942 and 0.775, which makes them a defensible alternative wherever a human-readable unit is preferred. Census tract fails for two compounding reasons: the grid is extremely fine, with mean demand of 2.1 trips per window, and, as established in Section 2, only 45.6% of trips carry a tract identifier while the suppressed ones are precisely the sparse observations. The unit is therefore unsuitable, not because the models are weak but because the input is simultaneously too fine and structurally incomplete in a demand-correlated way. Temporally the direction reverses, with R^2 rising from 0.927 at hourly to 0.958 at daily windows, simply because aggregation averages out noise. This is not an argument for daily buckets, since a daily figure cannot distinguish the vehicles needed at 8 a.m. from those needed at 6 p.m. Full tables for both sweeps are given in the Appendix.

Three limitations qualify these results. The first concerns the weather variables and is a direct consequence of the temporal resolution. Aggregating weather to four-hour windows attenuates intermittent conditions far more than smooth ones: temperature varies by only 3.2 °F within an average window and survives aggregation almost intact, whereas in windows recording rain only 58% of the constituent hours were actually wet, and in 35% of them the rain fell within a single hour. A window labeled as wet is therefore frequently dry for most of its duration, and its recorded total says nothing about whether the rain fell during its busiest or its quietest hour. This helps reconcile the small weight the weather terms carry in the model with the real association shown in Section 4.1: weather is a secondary driver to begin with, since rain shifts demand by 5 to 7% while the swing between night and late afternoon spans an order of magnitude, and aggregation blurs what remains. Finer bins would measure weather more faithfully but would make the demand panel too sparse to learn from, as the zero rate doubles from 12.6% to 26.8% at hourly resolution, so the trade-off cannot be resolved within a single resolution. Second, the target is served demand, so the models reproduce the historical supply-constrained pattern and understate demand wherever vehicles were unavailable. Third, the panel contains trip characteristics such as mean distance and fare that are known only after the trips have occurred. These are excluded to avoid leakage, at the cost of some predictive signal. The most promising improvements follow from these gaps: point-of-interest and event data to explain the residual variation around venues and airports, an explicit treatment of unmet demand, and a weather representation that preserves intensity within the window, for example the wet fraction of the window rather than its mean.

4.3 Prescriptive analytics: smart charging

Operating an electrified fleet introduces a recurring decision that a conventional fleet never faces: when, and how quickly, each vehicle is recharged. The decision repeats every day for every vehicle, so even small improvements compound across a fleet, which makes it a natural candidate for automation and a fair test of whether a learning agent can outperform a simple operating rule. We examine this on a representative case. A driver returns home at 2 p.m. and departs again at 4 p.m., leaving a two-hour window in which the vehicle can be charged at a variable rate before the next shift. Charging early and at full power guarantees a full battery but pays the highest prices, while charging sparingly is cheap but risks the vehicle running out of energy mid-shift. We divide the window into eight fifteen-minute decision slots and describe it as a Markov decision process. The state is the pair of slot index and current charge, the action is one of four throttle settings of a 22 kW wallbox (0, 7, 14 or 22 kW), and the transition is deterministic, adding the chosen power over a quarter of an hour up to the battery capacity of 60 kWh. The reward combines two terms: each slot incurs a charging cost convex in power, `alpha_t * (exp(0.1 * p) - 1)`, with `alpha_t` following an intraday price profile that makes later slots dearer, and at departure the daily energy requirement is drawn from a truncated normal distribution with mean 21.4 kWh and standard deviation 14.1 kWh, derived from the trip data at 0.40 kWh per mile. If that requirement exceeds the available charge, the day ends with a penalty of 20 plus 9.54 for every missing kilowatt-hour, the latter standing for the revenue lost on trips that cannot be served. The environment is implemented as a discrete-event simulation that receives an action and returns the next state and reward.

One modeling decision separates a planning exercise from a genuine learning problem. If every day is treated independently and always begins from the same charge, the optimal solution is a single fixed schedule that fills the battery as far as the window allows, and no state-dependent behavior is needed. We therefore couple consecutive days, so that whatever charge remains after a shift becomes the starting state of the next day. Because that starting charge now varies with yesterday's demand, the best action depends on the state rather than on the clock alone. The resulting schedules show this directly: a day beginning near empty charges at full power in all eight slots, whereas a day that inherits a substantial reserve throttles back and shifts the little it still needs into the cheaper early slots, for instance to `[22, 14, 7, 14, 22, 7, 7, 7]` kW.

We solve the problem twice, and the contrast is itself informative. Tabular Q-learning is the model-free learner: it knows neither the demand distribution nor the price profile and improves its policy purely from experienced rewards, over 200,000 simulated days with exploration decaying from 1.0 to 0.02. We deliberately use a table rather than a deep Q-network, because the state is two-dimensional and already discretized, so a lookup table represents the value function exactly and transparently, whereas a function approximator would introduce estimation variance without enlarging the space of representable policies. Dynamic programming provides the model-based counterpart, computing the optimal policy by value iteration from the known demand distribution and converging in 187 iterations. It plans rather than learns, and therefore serves as the yardstick for the learned policy, subject to the caveat that it discretizes demand on an 81-point grid and is thus an accurate reference rather than a proof of exact optimality.

Comparing policies requires care about the quantity being measured, since the agent optimizes charging cost and shortfall penalty jointly, and reporting charging cost alone would capture the smaller half of that objective for any policy that accepts risk. Table 4 therefore shows the total cost together with its components over 5,000 simulated days under an identical demand sequence, sorted from best to worst, where total cost is the sum of the charging and penalty columns and the deficit column averages only over days on which a shortfall occurs. Ranking by charging cost alone would reverse much of that order, as Figure 3 makes visible: plotting each policy's charging cost against its shortfall rate, only the planner occupies the low-cost, low-risk corner. It dominates every alternative at less than half the total cost of the best heuristic, charging cheaply while stranding the vehicle on only 0.5% of days. The learned policy does not succeed: it is beaten by two of the three naive heuristics on the objective it was trained to optimize. The diagnostic columns explain why. Q-learning achieves a low charging cost, second only to the planner, but strands the vehicle on 3.0% of days, and when it does the shortfall averages 8.2 kWh against 5.5 kWh under the planner. It has learned to economize on energy without learning how much reserve that economy requires. The table also shows how misleading charging cost alone would be, since charging always at maximum power appears by far the worst policy on cost yet ranks third on total cost because it almost never strands. We read the gap between the two methods as the price of not knowing the model: a single penalty at the end of the day must be attributed back across eight decisions and then across the day boundary, and 200,000 days of experience are evidently not enough to resolve that credit assignment.

Table 4: Charging policy performance over 5,000 simulated days.

Policy	**Charging**	**Shortfall**	**Deficit**	**Penalty**	**Total**
Dynamic programming	1.05	0.5%	5.5 kWh	0.38	1.43
Cheapest slots	2.11	1.4%	6.8 kWh	1.22	3.33
Always maximum	3.37	0.5%	5.6 kWh	0.34	3.71
Q-learning	1.32	3.0%	8.2 kWh	2.96	4.28
Constant medium	1.28	3.9%	7.9 kWh	3.71	4.99

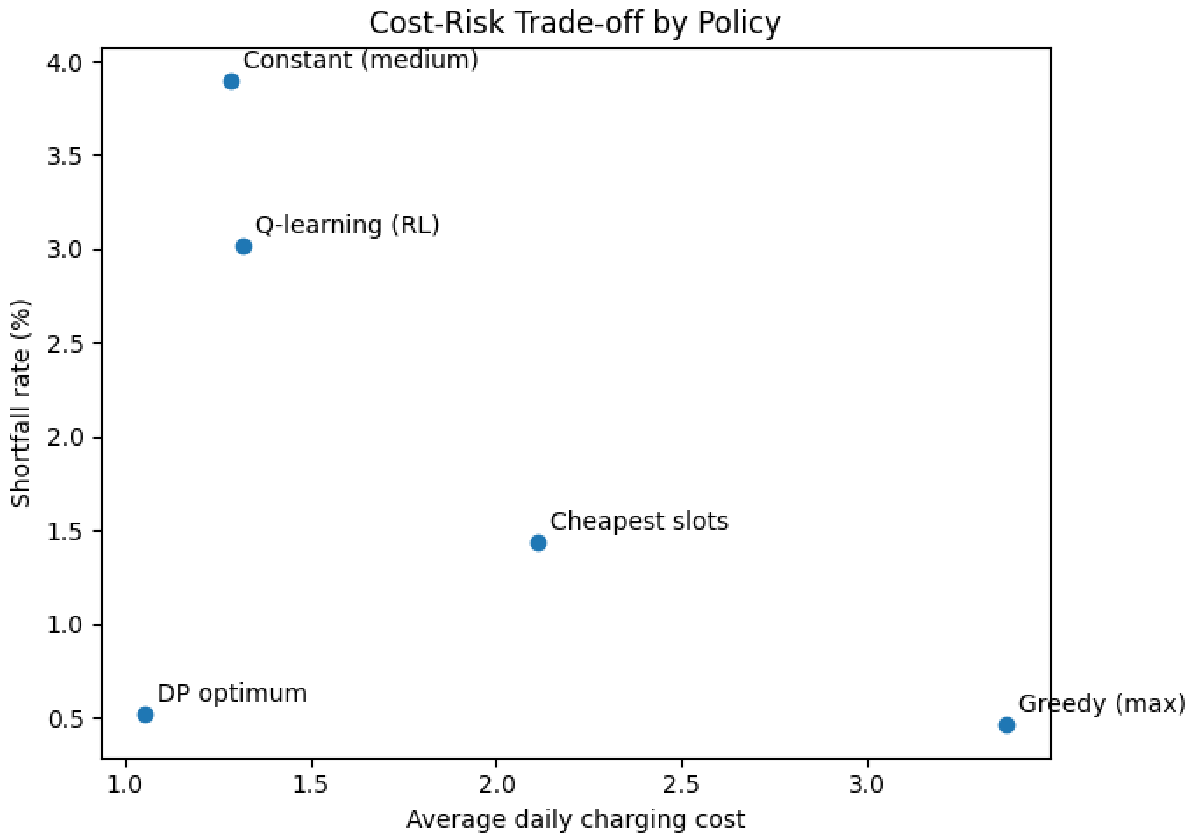

Figure 3: Average daily charging cost against shortfall rate for each policy.

Sensitivity analysis identifies which assumptions carry the result. Because the planner is re-solved for every setting, these figures show how the attainable optimum shifts rather than how a fixed policy degrades. Demand variability dominates: raising the standard deviation from 5 to 20 kWh lifts the optimal shortfall rate from 0.1% to 3.6%, so the residual risk is a property of demand uncertainty rather than of the charging policy. Battery capacity is the second lever, since 40 kWh leaves an irreducible 9.2% shortfall rate that 60 kWh reduces to 0.2% and 80 kWh removes entirely, with the larger pack also lowering charging cost because a greater reserve permits more shifting into cheap slots. The parameter we chose most freely matters least: multiplying both penalty components by eight moves the optimal shortfall rate only from 0.45% to 0.25%, so the conclusions do not rest on the assumed cost of stranding. These figures are outcomes of a stylized model rather than monetary forecasts. They depend on the assumed energy conversion, the truncated normal demand, the synthetic price profile and the fixed charging window, and they omit charging losses, battery degradation, public charging and any interaction between vehicles in a fleet.

5 Conclusions and recommendations

Three findings carry this project. Demand for on-demand mobility in Chicago is strongly and stably structured in space and time, concentrating in a few cores that behave differently from one another: a dense downtown producing many short trips, and airport zones producing fewer but longer and more valuable ones. That structure is learnable, since framing demand as a function of calendar, location and weather rather than of its own recent history yields a model that beats a naive historical lookup, although only the neural network manages this. Charging, finally, proved to be governed less by the vehicle than by how long it stands still.

Methodologically, three **strengths** stand out. The pipeline is transferable, converting raw, privacy-restricted trip records into a clean, weather-enriched demand panel, with nothing specific to Chicago beyond the input file, so it can be re-run for whichever city the operator enters. The choice of spatial resolution rests on three mutually independent criteria, namely statistical sparsity, the measurement validity implied by centroid-based coordinates, and the agreement between grid-free mixture components and grid cells. All three deteriorate at the same step, which makes the resulting unit well supported rather than convenient. Most importantly, benchmarking against trivial alternatives was decisive rather than decorative: comparing against a historical-mean lookup revealed that a fully tuned kernel machine does not beat it, which prevented us from recommending a model that would have looked respectable in isolation. The same discipline applied to charging, where solving the problem both with a model-based planner and with a model-free learner made the cost of not knowing the model visible, which either method alone would have concealed.

Equally clear are the **limitations**, and each points to the further analysis or the external data that would resolve it. The records capture served demand only, so the models reproduce a supply-constrained history and understate demand wherever vehicles were unavailable. The remedy lies within the existing data: trip durations, idle times and vehicle identifiers allow vehicle productivity to be estimated, which is also the missing step from predicted demand to required fleet size. A fleet simulation over the observed origin-destination flows would further show where vehicles accumulate and how much empty repositioning is needed. Taxis remain a proxy that differs from app-based hailing in booking behavior and market coverage. Fine spatial units are unusable because the privacy suppression correlates with demand, a transferable lesson rather than a local quirk: an operator entering any US city should not assume fine-grained open data can be taken at face value. We can describe where demand differs but not why neighboring cells do, and the external data that would close this gap are point-of-interest records for offices, transit stations, hotels and venues, together with event calendars. Robustness also deserves more scrutiny than average accuracy provides, since planning depends on how badly a model fails at peaks rather than on typical-day performance. Rolling time-based validation, prediction intervals and residual analysis would establish this. The charging study is the most stylized component. Its two-hour afternoon window is an assumption rather than an observation, and a real fleet would charge overnight or opportunistically at low-price hours, so the constraint we identified is partly an artifact of that setup. The study also uses a synthetic price profile, ignores charging losses and degradation, and models one vehicle rather than a fleet. It should therefore be read as a controlled proof of concept requiring telematics, real tariffs and public charger data, including utilization, waiting times and availability, before informing an infrastructure decision. It produced our clearest negative result, since the reinforcement-learning agent failed to match simple heuristics on its own objective, which points to a larger training budget or function approximation rather than to a flaw in the formulation.

For the operator, our **recommendations** follow the three decision levels set out at the outset. Strategically, market entry should begin with the persistent demand cores and the corridors connecting them rather than with city-wide coverage, since these are the densest and most reliable sources of demand. Tactically, deployment should be planned per shift and per area rather than as a flat fleet number, while recognizing that the model predicts trips and not vehicles, a conversion requiring the productivity estimate described above. Operationally, repositioning should target the corridors with the strongest directional asymmetry, and charging should be scheduled into the periods and locations where vehicles already stand idle, which our idle-time analysis identifies directly. On infrastructure, the results favor private charging as the operational backbone, with public charging as targeted insurance rather than a substitute. Private charging is cheaper per kilowatt-hour and controllable in timing, but its capacity is bounded by how long vehicles stand still, whereas public charging costs more yet is available when they do not. The decisive lever is therefore scheduling charging into long idle periods, typically overnight, rather than optimizing power within a short window. Public capacity should be procured only against the residual tail of days on which demand exhausts what those idle periods can deliver. A pilot in the target market should calibrate how thick that tail is and how reliable local public charging proves to be.

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A Appendix

A.1 Field dictionary of the raw trip records

The main text groups the 23 raw fields by type for readability. The table below resolves that grouping into the individual fields, giving for each its name as published, a short description, and its data type, so that no question about the source data remains open.

Table 5: Field dictionary of the raw trip records

Column Name	Description	Data Type
Trip ID	A unique identifier for the trip	Text
Taxi ID	A unique identifier for the taxi	Text
Trip Start Timestamp	When the trip started, rounded to the nearest 15 minutes	Floating Timestamp
Trip End Timestamp	When the trip ended, rounded to the nearest 15 minutes	Floating Timestamp
Trip Seconds	Time of the trip in seconds	Number
Trip Miles	Distance of the trip in miles	Number
Pickup Census Tract	The Census Tract where the trip began	Text
Dropoff Census Tract	The Census Tract where the trip ended	Text
Pickup Community Area	The Community Area where the trip began	Number
Dropoff Community Area	The Community Area where the trip ended	Number
Fare	The fare for the trip.	Number
Tips	The tip for the trip. Cash tips generally will not be recorded.	Number
Tolls	The tolls for the trip.	Number
Extras	Extra charges for the trip.	Number
Trip Total	Total cost of the trip, the total of the previous columns.	Number
Payment Type	Type of payment for the trip.	Text
Company	The taxi company.	Text

Column Name	Description	Data Type
Pickup Centroid Latitude	The latitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy	Number
Pickup Centroid Longitude	The longitude of the center of the pickup census tract or the community area if the census tract has been hidden for privacy	Number
Pickup Centroid Location	The location of the center of the pickup census tract or the community area if the census tract has been hidden for privacy	Point
Dropoff Centroid Latitude	The latitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy	Number
Dropoff Centroid Longitude	The longitude of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy	Number
Dropoff Centroid Location	The location of the center of the dropoff census tract or the community area if the census tract has been hidden for privacy	Point

A.2 Validation rules and thresholds

Section 3 distinguishes records that cannot describe a real journey, which are excluded outright, from those implausible in a single attribute, which are retained as demand events but withheld from the statistics they would distort. The table below lists every rule applied, the number of records affected and whether the rule acts as a hard exclusion or a soft one.

Table 6: Validation rules and thresholds

Flag	Description	Scope	Rows	Percentage
end_before_start	Trip end timestamp is earlier than the trip start timestamp	hard — all aggregations	98	0.000662%

Flag	Description	Scope	Rows	Percentage
extreme_seconds	Trip duration exceeds 3 hours (10,800 s)	hard — all aggregations	14438	0.097554%
extreme_miles	Trip distance exceeds 100 miles	hard — all aggregations	1202	0.008120%
invalid_seconds	Trip duration is zero	soft — duration/speed features only	201927	1.364371%
zero_miles	Trip distance is recorded as zero, likely reflecting metering issues or canceled trips; still counted as a demand event.	soft — distance/speed features only	1294820	8.747182%
extreme_total	Trip total exceeds $200; mostly plausible long-distance rides, but a subset lacks a plausible distance.	soft — revenue aggregations only	9541	0.064598%
missing_pickup_area	Pickup community area is missing (mainly trips outside Chicago).	spatial — community area datasets only	413385	2.792604%

A.3 Spatial resolution reference

Table 7: Average cell area of each H3 resolution with an intuitive scale equivalent.

Resolution	Average area	Rough scale
r4	1,770.3 km^2	Large city or metro region
r5	252.9 km^2	City district
r6	36.1 km^2	Large neighborhood
r7	5.2 km^2	Neighborhood
r8	0.7 km^2	A few city blocks

A.4 Descriptive analytics

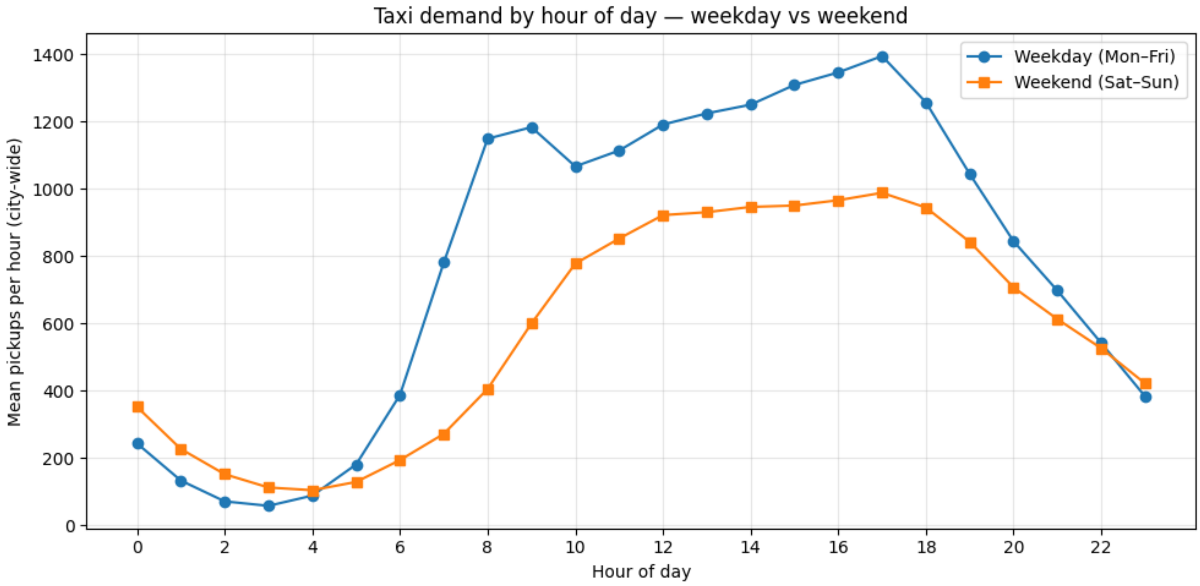


Figure 4: Taxi demand by hour of day, split into weekdays and weekends. The line view complements the heatmap in Section 4.1 by showing the shape of the two profiles directly against one another.

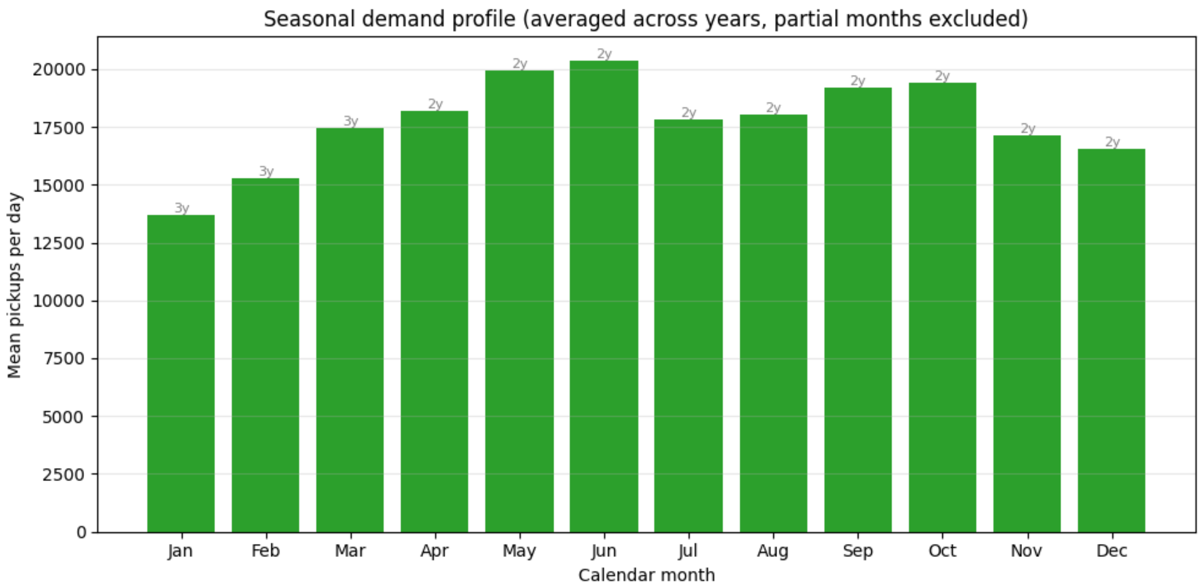


Figure 5: Seasonal demand profile across the observation period, showing the amplitude between the quiet winter months and the summer peak.

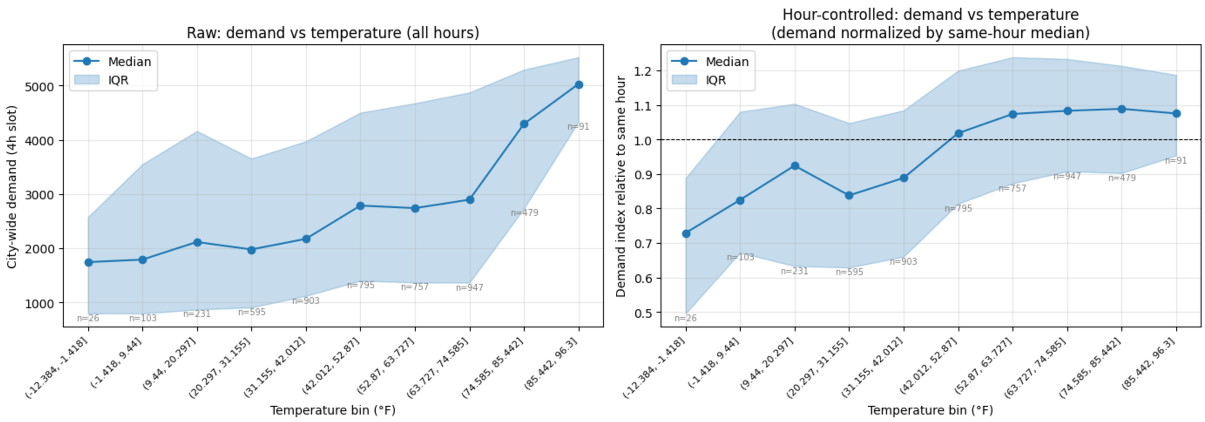


Figure 6: Demand against temperature, aggregated to four-hour windows. The association remains when windows are compared within the same time of day.

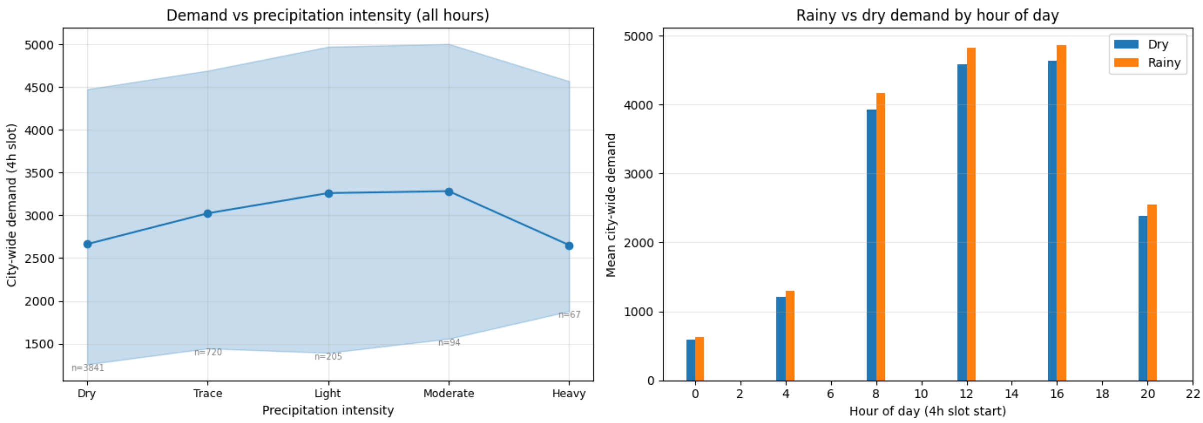


Figure 7: Demand against precipitation intensity, aggregated to four-hour windows.

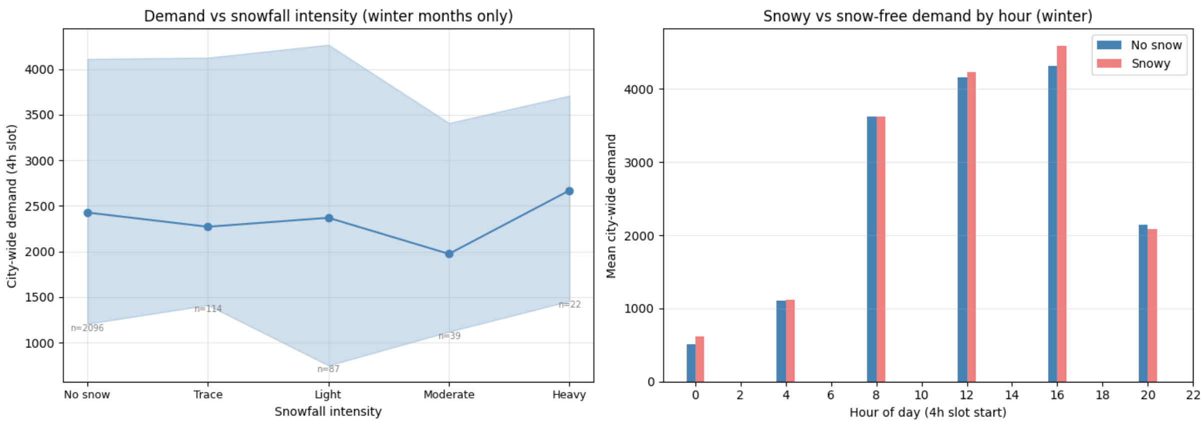


Figure 8: Demand against snowfall intensity, restricted to winter months. The windows with heavy snowfall are too few to support a conclusion, which is why Section 4.1 reports no reliable effect.

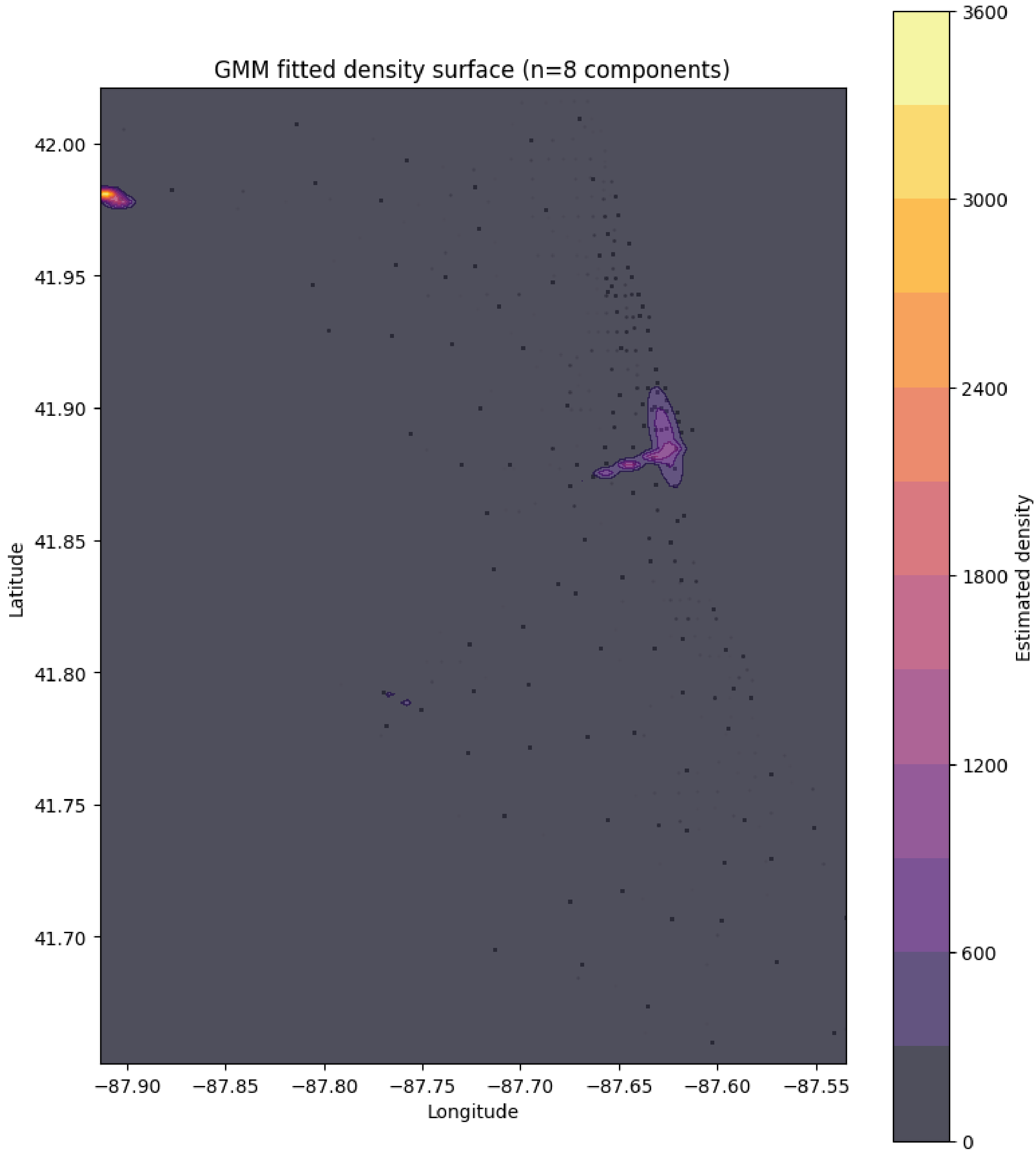


Figure 9: Gaussian mixture model fitted to the raw pickup coordinates. Because the model uses no grid, its component centers identify demand cores independently of the hexagon discretization.

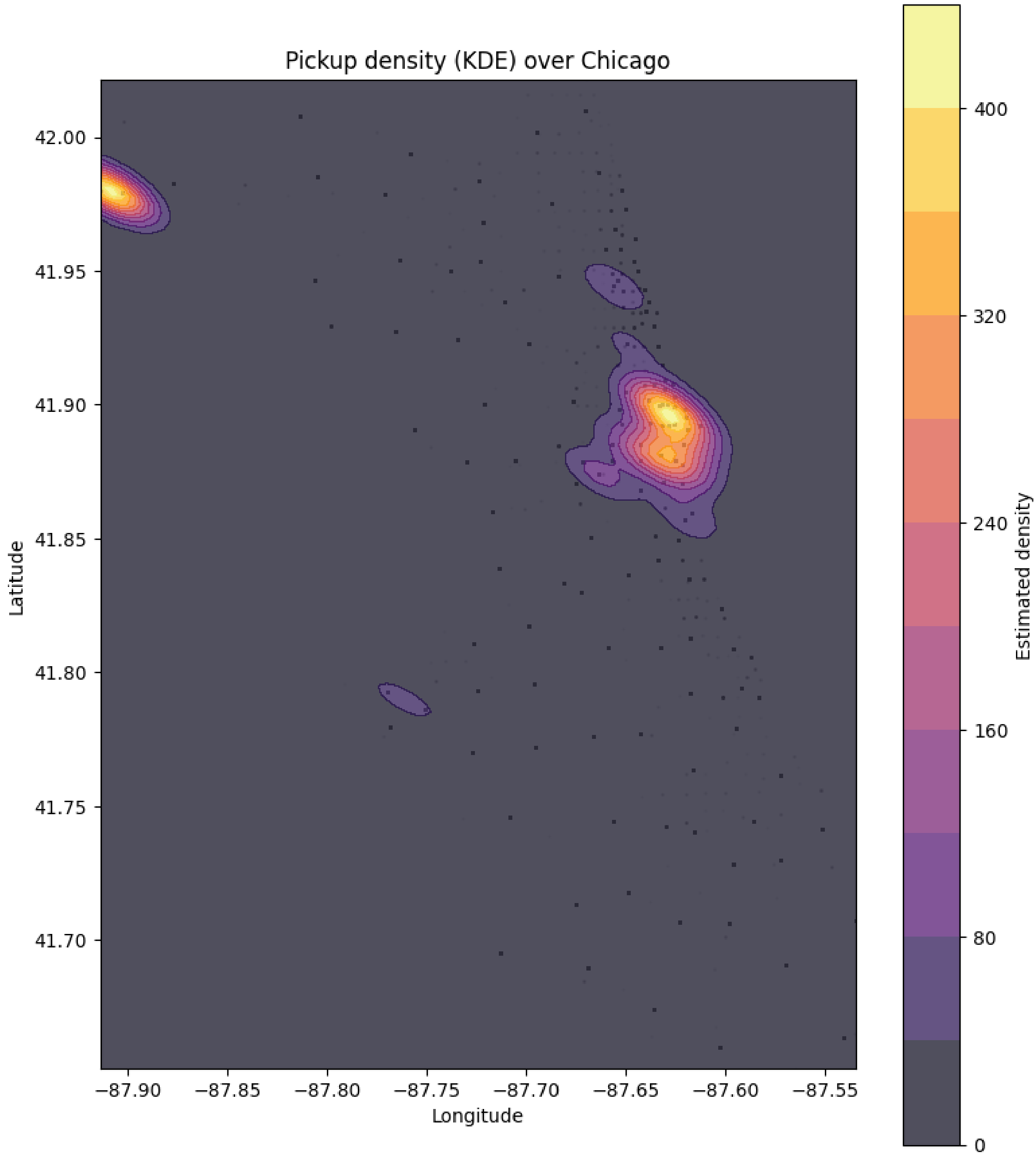


Figure 10: Kernel density estimate of pickup density. Unlike the mixture components, it shows the continuous density gradient rather than discrete centers.

A.5 Feature definitions

The predictive models use eighteen features. The table below defines each one, stating how it is derived from the demand panel, in particular the cyclical encoding of the calendar quantities and the great-circle distances to the three demand centers.

Table 8: Feature definitions

Feature	**Group**	**Type**	**Meaning**
`demand_count`	target	int	trips originating in a cell × 4h bucket (prediction target)
`hour_sin` / `hour_cos`	calendar	float	time-of-day (4h buckets), cyclical encoding
`dow_sin` / `dow_cos`	calendar	float	day-of-week, cyclical encoding
`month_sin` / `month_cos`	calendar	float	month/season, cyclical encoding
`is_weekend`	calendar	int (0/1)	weekend flag
`cell_lat`	spatial	float	cell centroid latitude
`cell_lon`	spatial	float	cell centroid longitude
`dist_to_loop_mi`	spatial	float	miles to downtown core
`dist_to_ohare_mi`	spatial	float	miles to O'Hare
`dist_to_midway_mi`	spatial	float	miles to Midway
`temp_4h`	weather	float	avg temperature (°F)
`rain_4h`	weather	float	total rainfall accumulated
`snow_4h`	weather	float	total snowfall accumulated
`precip_4h`	weather	float	total precipitation accumulated
`wind_speed_4h`	weather	float	avg wind speed
`weather_code_4h`	weather	int (categorical)	max weather condition code

A.6 Resolution sensitivity results

Section 4.2 reports the direction of the resolution effects and the values at the decisive points. The two tables below give the complete sweeps, spatially across r5, r6, r7, and community areas at fixed four-hour windows, and temporally across one-hour, four-hour, eight-hour and daily windows at fixed H3 r6. Both report cell count, zero rate, mean demand, MAE, normalized MAE, RMSE and R^2 for each model. Note that these refits reuse the tuned configuration rather than repeating the search at every resolution, so they are comparable across resolutions but not directly against the holdout results in the main text.

Table 9: Resolution sensitivity results of spatial dimension

Spatial	Temporal	Model	n_cells	Zero Rate	Mean	MAE	nMAE	RMSE	R^2
h3_r5	4h	RBF SVM	8	0.010	356.676	62.077	0.174	149.005	0.937
h3_r5	4h	Tuned NN	8	0.010	356.676	61.391	0.172	139.062	0.945
h3_r6	4h	RBF SVM	27	0.126	105.682	26.144	0.247	99.349	0.913
h3_r6	4h	Tuned NN	27	0.126	105.682	21.051	0.199	77.970	0.946
h3_r7	4h	RBF SVM	128	0.492	22.292	15.792	0.708	79.403	0.449
h3_r7	4h	Tuned NN	128	0.492	22.292	8.810	0.395	42.321	0.843
community_area	4h	RBF SVM	77	0.173	37.040	17.285	0.467	78.056	0.702
community_area	4h	Tuned NN	77	0.173	37.040	10.497	0.283	44.611	0.903

Table 10: Resolution sensitivity results of temporal dimension

Spatial	Temporal	Model	n_cells	Zero Rate	Mean	MAE	nMAE	RMSE	R^2
h3_r6	1h	RBF SVM	27	0.268	26.424	7.876	0.298	28.822	0.888
h3_r6	1h	Tuned NN	27	0.268	26.424	6.598	0.250	23.340	0.927
h3_r6	4h	RBF SVM	27	0.126	105.696	26.432	0.250	100.433	0.911
h3_r6	4h	Tuned NN	27	0.126	105.696	21.810	0.206	80.674	0.942
h3_r6	8h	RBF SVM	27	0.078	211.392	46.269	0.219	174.723	0.926
h3_r6	8h	Tuned NN	27	0.078	211.392	38.193	0.181	138.284	0.954
h3_r6	1d	RBF SVM	27	0.040	634.177	122.432	0.193	413.329	0.938
h3_r6	1d	Tuned NN	27	0.040	634.177	103.228	0.163	341.921	0.958

A.7 Neural network specification and diagnostics

The tuning grid below is deliberately small, since the search runs inside grouped cross-validation and is repeated within each fold.

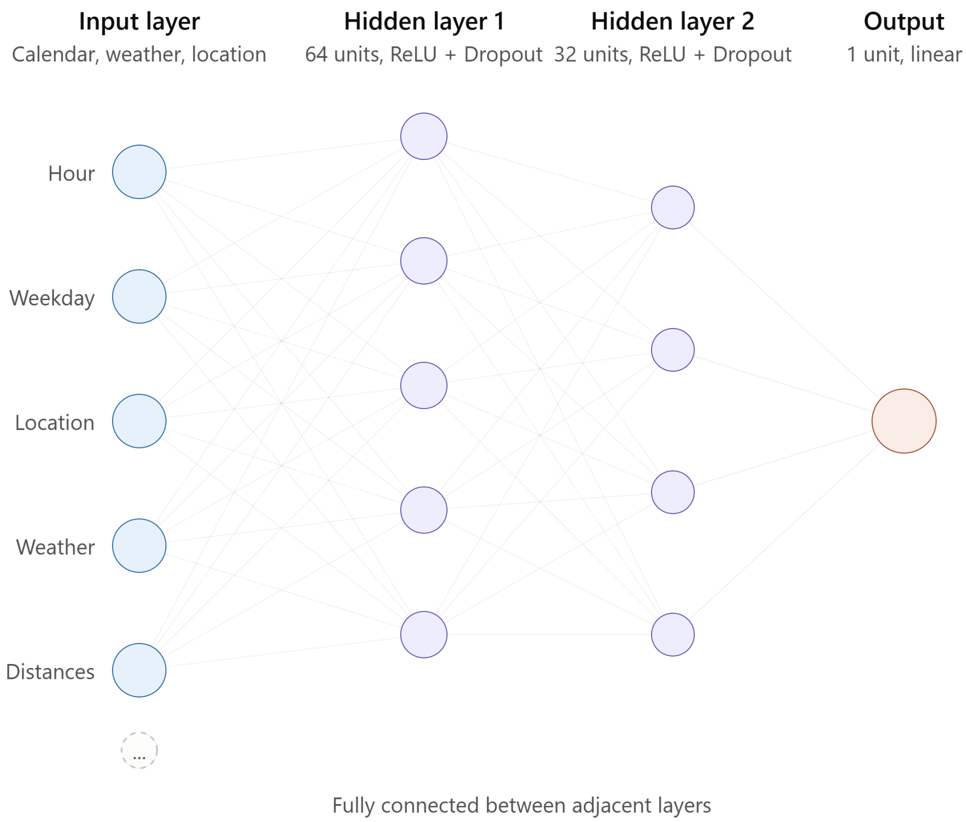


Figure 11: Architecture of the baseline neural network, showing layer sizes and the arrangement of the feedforward stack.

Table 11: Candidate values searched when tuning the neural network.

Parameter	**Candidate values**
`hidden_sizes`	(64,), (64, 32), (128, 64, 32)
`dropout`	0.1, 0.2
`lr`	3e-3, 1e-3, 5e-4
`batch_size`	256, 512

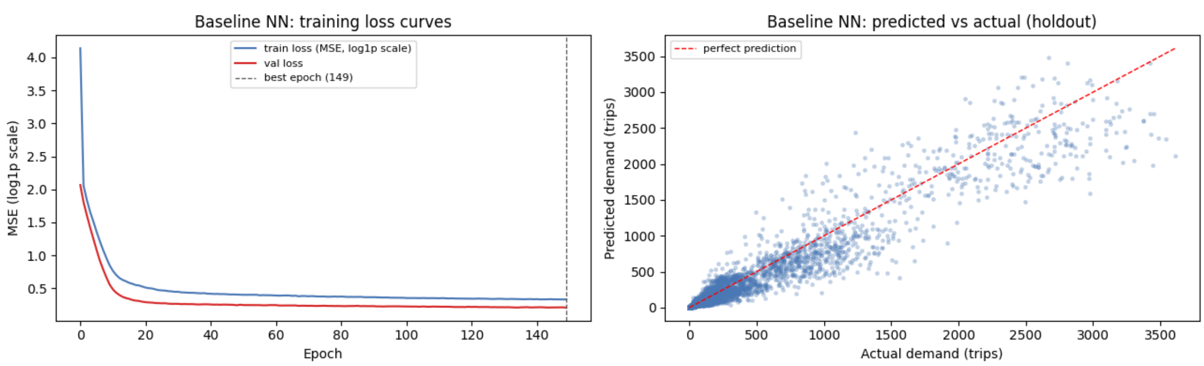
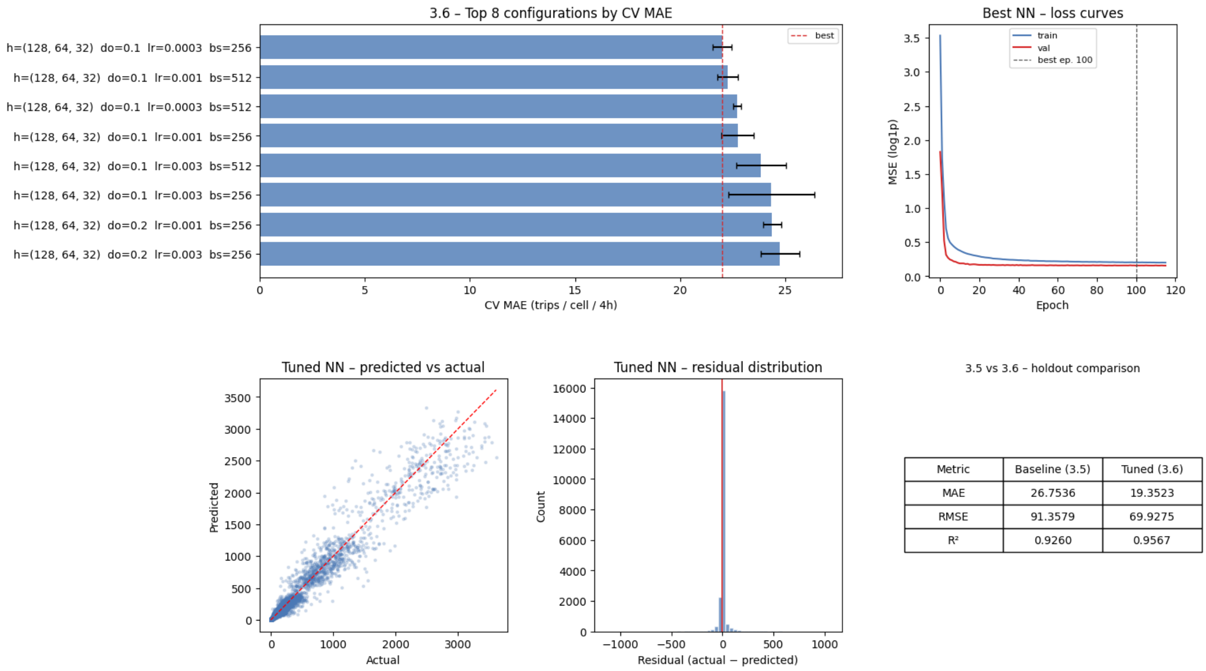


Figure 12: Training result of the baseline neural network, before hyperparameter tuning.



Metric	Baseline (3.5)	Tuned (3.6)
MAE	26.7536	19.3523
RMSE	91.3579	69.9275
R²	0.9260	0.9567

Figure 13: Training result of the tuned neural network, the configuration carried into the holdout comparison.

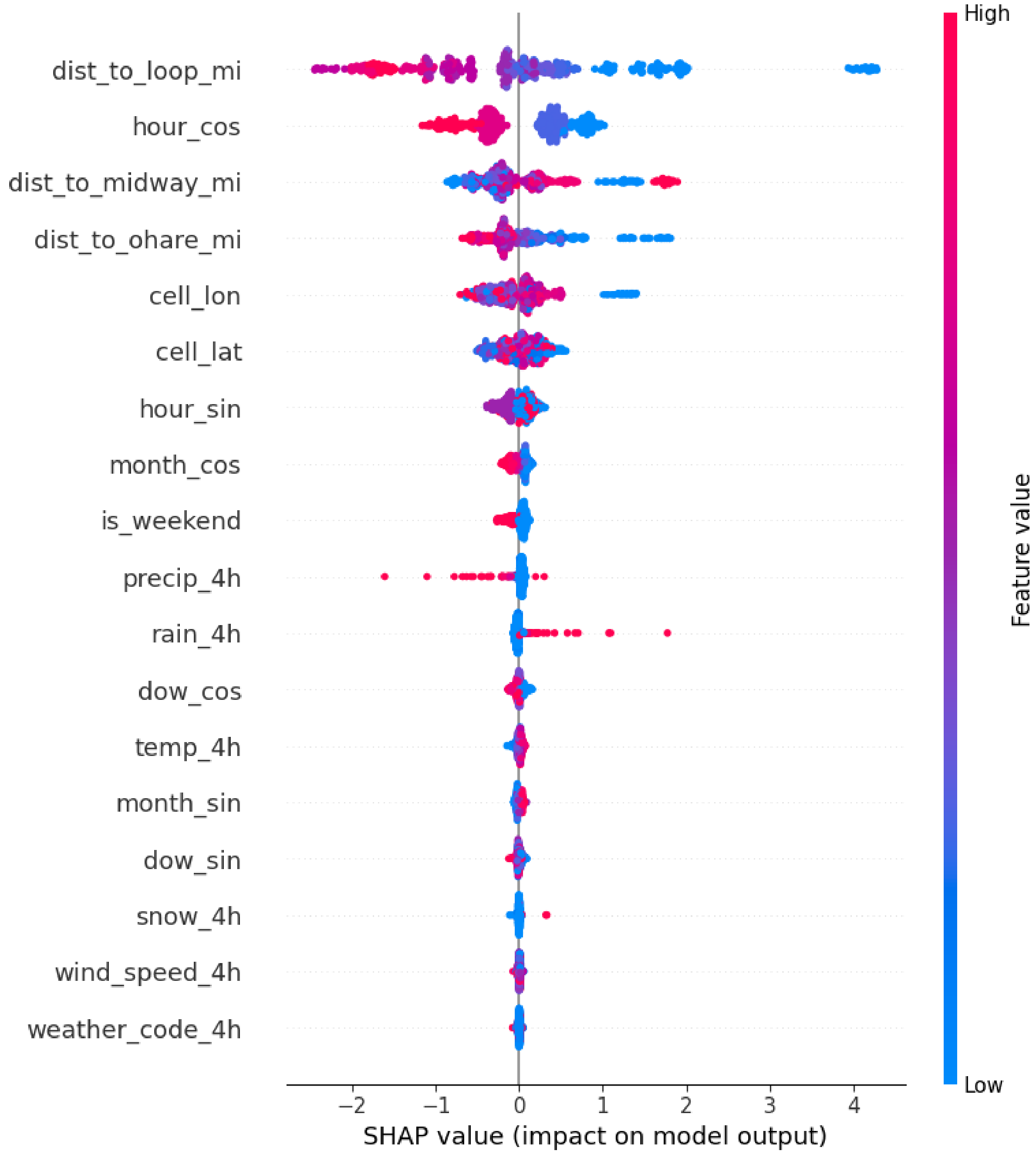


Figure 14: SHAP feature importance for the tuned neural network, showing the distances to the demand centers and the time-of-day terms dominating the prediction.